



Answers

Extending thinking about sequences

Starter quiz

© Oak National Academy 2024. Produced in partnership with Mathematics in Education and Industry.

Licensed on the [Open Government Licence v3.0](#), except where otherwise stated. [See Oak terms & conditions](#).



0

Extending thinking about sequences

1 An arithmetic (or linear) sequence is a sequence where the difference between successive terms is a _____. Tick **1** correct answer

- ☒ constant
- ☐ multiplier
- ☐ negative number
- ☐ positive number
- ☐ variable

2 Find the next term in the arithmetic sequence -25, -18, -11, -4, 3. Fill in the blank

3 Find the missing term in the arithmetic sequence 27, 41, 55, 69, 83, ... Fill in the blank

4 Find the common ratio of the geometric sequence 0.4, 1.6, 6.4, 25.6, ... Tick **1** correct answer

- ☐ 0.25
- ☐ 0.4
- ☐ 0.6
- ☒ 4
- ☐ 6

5 Find the next term in the geometric sequence 0.4, 1.6, 6.4, 25.6, 102.4, ... Give your answer as a decimal. Fill in the blank

6 Which statements are true of the sequence 1920, 960, 480, 240, 120, ... ? Tick **2** correct answers

- ☐ It is a decreasing sequence therefore will contain negative values in the future
- ☐ It is a geometric sequence with a common ratio greater than 1
- ☒ It is a geometric sequence with a common ratio less than 1
- ☐ It will eventually start to increase again.
- ☒ It will never reach zero.